

Dheeraj Verma

Machine Learning Engineer | Applied AI | MLOps

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Summary

Machine Learning Engineer with hands-on experience in end-to-end ML pipelines, MLOps (CI/CD, Docker, AWS), NLP, and Generative AI using Transformers and RAG systems.

Education

Jaypee University of Engineering and Technology , B.Tech Computer Science and Engineering 2022-2026

Micro-specialization in Artificial Intelligence and Machine Learning

Experience

Prodigy Infotech | Generative AI Intern | <https://prodigyinfotech.dev/> 8/2025 – 9/2025

- Implemented an **image-to-image translation model** using **conditional GAN (pix2pix)** for paired image datasets.
- Engineered and trained **generator–discriminator architecture** in **TensorFlow/Keras**, achieving high-quality image outputs.
- Applied **data preprocessing, loss optimization, and training stabilization techniques** to improve model accuracy and stability.
- Gained hands-on expertise in **GAN-based architectures, adversarial training, and computer vision** applications.

Proxenix | Data Science and Analytics Intern | <https://proxenix.com/> 4/2025 – 6/2025

- Developed document summarizer using **BART transformer**, reducing document reading time by 70%.
- Designed **NLP pipelines** for large datasets with **tokenization, embeddings, and model optimization**.
- Experimented with transformer **fine-tuning** and evaluation metrics (**ROUGE, BLEU**) for summarization quality.

Projects

YouTube Sentiment Insights – MLOps Project | [GitHub](#)

- Developed end-to-end **CI/CD pipeline** (GitHub Actions, MLflow, DVC, Docker, AWS) automating builds, testing, and deployment.
- Configured **Docker containers** for reproducible environments and integrated version control with Git.
- Optimized LightGBM with **TF-IDF + SMOTE** achieving ~80% accuracy.

AI Fitness Assistant – Multi-Agent ML Project

- Built a **Multi-Agent RAG-powered AI fitness** assistant using **Python, Streamlit, and AstraDB**.
- Designed modular **multi-agent architecture** (nutrition, fitness Q&A, calculator) with **Retrieval-augmented generation (RAG)**, improving relevance by ~80% and reducing compute costs by ~40%.

Skills & abilities

- Languages & CS Foundations:** Python, C++, SQL, Data Structures & Algorithms, OOP, System Design (basics)
- Software Development:** Git, GitHub, CI/CD (GitHub Actions), Docker, Linux, Agile/Scrum
- AI/ML :** NumPy, Pandas, Scikit-learn, Transformers, Hugging face, LangChain, LangGraph, Encoder-Decoder Architecture, Machine Learning, NLP, Deep Learning

Position of Responsibility and Certifications

Google Developer Student Club - Coordinator 9/2023 – 8/2024

Part of Student body mentored 50+ students during interactive sessions.

Oracle Cloud Infrastructure 2025 AI Foundations Associate (1Z0-1122-25) 9/2025 – 9/2027

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional 10/2025 – 10/2027